BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 6191 / 6194

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6191

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Multiplicative inverse of - i is:

(A) i
(B) - i
(C) 1
(D) - 1

Q2. If n is prime, then √n is:

(A) Rational number
(B) Whole number
(C) Natural number
(D) Irrational number

Q3. A function f: A → B is surjective if:

(A) Range f = A
(B) Range f = B
(C) Range f ≠ A
(D) Range f ≠ B

Q4. Set of integers is a group with respect to:

(A) +
(B) ÷
(C) ×
(D) -

Q5. Which symbol is used for membership of a set?

(A) $\wedge$ (B) $\vee$

(C) $\in$ (D) $\sim$

Q6. Transpose of diagonal matrix is:

(A) Scalar matrix (B) Row matrix

(C) Null matrix (D) Diagonal matrix

Q7. For any non-singular matrix A, A^{-1} is:

(A) $|A|$ adj (A) (B) $1/(|A|$ adj (A))

(C) adj (A)/$|A|$ (D) $|A|$/adj (A)

Q8. A quadratic equation $ax^2 + bx + c = 0$ becomes linear equation if:

(A) $a = 0$ (B) $b = 0$

(C) $c = 0$ (D) $a = b$

Q9. If ω is complex root of unity then value of $(3 + \omega)(3 + \omega^2)$ is:

(A) 6 (B) 7

(C) 9 (D) 13

Q10. The improper fraction can be changed into proper fraction by:

(A) Addition (B) Subtraction

(C) Multiplication (D) Division

Q11. The sequence 3, 6, 12 is:

(A) A.P (B) G.P

(C) H.P (D) Arithmetic series

Q12. If a, b are negative and G.M is also negative then:

(A) $H < A < G$ (B) $A < H < G$

(C) $G < A < H$ (D) $A < G < H$

Q13. If n is a negative integer then $\underline{n}$ is:

(A) 1 (B) Not defined

(C) Zero (D) n

Q14. The number of term in the expansion of $(a + b)^n$ is:

(A) $n^2 + 1$ (B) $n + 1$

(C) $n - 1$ (D) n

Q15. The 60th part of 1 degree is called one:

(A) Second
(B) Radian
(C) Degree
(D) Minute

Q16. $\cot(\pi - \alpha) =$:

(A) $\sin \alpha$
(B) $\cot \alpha$
(C) $-\cot \alpha$
(D) $\tan \alpha$

Q17. The domain of cosx is:

(A) [-1,1]
(B) $(-\pi/2, \pi/2)$
(C) R
(D) Q

Q18. A circle passing through the three vertices of a triangle is called:

(A) Circumcircle
(B) In-circle
(C) Escribed circle
(D) Both A and B

Q19. $\sin^{-1}(-1/2) =$:

(A) $\pi/3$
(B) $-\pi/6$
(C) $\pi/4$
(D) $-\pi/3$

Q20. If sinx = cosx, then x = :

(A) 45°
(B) 30°
(C) 0°
(D) 60°

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 1109-XI124

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) State Golden rule of fractions and rule for quotient of fractions.

(ii) Find multiplicative inverse of $(\sqrt{2}, \sqrt{5})$

(iii) Prove that sum as well as product of two conjugate complex numbers is a real.

(iv) Simplify: $(a + bi)^{-2}$

(v) Write power set of A = {9, 11}

(vi) If a, b are elements of a group G, then show $(ab)^{-1} = b^{-1}a^{-1}$

(vii) If A = [i 1+i ; 1 -i], show that A - $(\bar{A})^t$ is skew-Hermitian.

(viii) Evaluate: | 1 2 -3 ; -1 3 4 ; -2 5 6 |

(ix) If $A = [\ 2i\ i\ ;\ i\ -i\ ]$ then find A^{-1}

(x) Find k if $x^3 + kx^2 - 7x + 6$ has remainder - 4, when divided by x + 2.

(xi) α ,β are roots of $5x^2 - x + 2 = 0$, find $3/\alpha + 3/\beta$

(xii) Discuss the nature of roots of $x^2 - 5x + 6 = 0$

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Define rational fraction.

(ii) Resolve into partial fractions $(x^2 + x + 1)/(x + 2)^3$

(iii) Define sequence.

(iv) If the 5th term of an AP is 13 and its 17th term is 49, find its general term.

(v) Find vulgar fraction equivalent to $1.5\overline{3}$ recurring decimal.

(vi) Find the 12th term of harmonic sequence 1/3, 2/9, 1/6, ...

(vii) Evaluate: 10!/7!

(viii) Define sample space.

(ix) Find the value of n if ${}^nC_{10} = 12\times11/2!$

(x) Show that 5^n -1 is divisible by 4 if n = 5.

(xi) Expand (1 - 2x)^(1/3) up to 4 terms.

(xii) Find the middle term of $(x/2 + 2/x^2)^{12}$

Q4. (Marks 18) Attempt any NINE parts:

(i) If $\cos\theta = -\sqrt{3}/2$ and the terminal arm of the angle is in III quadrant, find the value of $\sin\theta$ and $\tan\theta$

(ii) Verify that $\cos2\theta = 2\cos^2\theta - 1$ when $\theta = 30°$

(iii) Show that $\cos(\alpha + \beta)\cos(\alpha - \beta) = \cos^2\beta - \sin^2\alpha$

(iv) Prove that $\cot\alpha - \tan\alpha = 2\cot2\alpha$

(v) Prove that $(\sin3x - \sin x)/(\cos x - \cos3x) = \cot2x$

(vi) What is the domain and range of $y = \cos x$?

(vii) Find the period of $3\cos(x/5)$

(viii) Draw the graph of $y = \sin x$ for $0 \le x \le 360°$.

(ix) Find the measure of the greatest angle if sides of triangle are 16, 20, 33.

(x) Find the area of the triangle ABC, when b = 37, c = 45, $\alpha = 30°50'$

(xi) Prove that $r_1r_2r_3 = rs^2$

(xii) Prove that $2\tan^{-1}A = \tan^{-1}(2A/(1 - A^2))$

(xiii) Find the solution of equation $\sec x = -2$, $x \in [0, 2\pi]$

Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6194

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If $\sin x = 1/2$, then $x =$:

(A) $\pi/6, 5\pi/6$ (B) $-\pi/6, 5\pi/6$

(C) $-\pi/6, -5\pi/6$ (D) $\pi/3, 2\pi/3$

Q2. Domain of $y = \tan^{-1}x$ is:

(A) R (B) Q

(C) N (D) Z

Q3. Angle below the horizontal line is called:

(A) Right angle (B) Oblique angle

(C) Angle of depression (D) Angle of elevation

Q4. Period of $\cos\theta$ is:

(A) 2π (B) $3\pi/2$

(C) π (D) $\pi/2$

Q5. $\cos(\theta - 90°) - \cos(\theta + 90°)$ equals to:

(A) $-2\cos\theta$ (B) $2\cos\theta$

(C) $-2\sin\theta$ (D) $2\sin\theta$

Q6. The angle $\pi/12$ in degree measure is:

(A) 30° (B) 20°

(C) 45° (D) 15°

Q7. If $n^2 > n + 3$ then it is true for:

(A) $n = 0$ (B) $n < 1$

(C) $n \geq 2$ (D) $n \geq 3$

Q8. For an event A, the range of P(A) is:

(A) $0 < P(A) \leq 1$ (B) $0 \leq P(A) < 1$

(C) $0 \leq P(A) \leq 1$ (D) $0 < P(A) < 1$

Q9. If $a = 1$, $b = 5$ then $A \times H$ is equal to:

(A) 5 (B) -5

(C) -5/2 (D) 2/5

Q10. Sum of infinite geometric series is valid if:

(A) $r < 1$ (B) $|r| < 1$

(C) $|r| = 1$ (D) $|r| > 1$

Q11. The fraction $(x + 1)/(x^2 + 2)$ is:

(A) Proper fraction (B) Improper fraction

(C) Identity (D) Mixed

Q12. Complex cube roots of -1 are :

(A) ω, ω^2 (B) $1, \omega, \omega^2$

(C) $-1, -\omega, -\omega^2$ (D) $-\omega, -\omega^2$

Q13. Sum of all the three cube roots of unity is:

(A) 1 (B) -1

(C) 3 (D) 0

Q14. The additive inverse of matrix A is:

(A) A (B) -A

(C) A^2 (D) 1

Q15. The trivial solution of homogeneous linear equations is:

(A) (1, 0,0) (B) (0, 1, 0)

(C) (0, 0, 1) (D) (0, 0, 0)

Q16. The domain of f = {(a, 1), (b, 1), (c, 1)} is:

(A) {a, b, c} (B) {1}

(C) {b, c} (D) {a, b, c, 1}

Q17. A function which is onto is called:

(A) Injective (B) Surjective

(C) Objective (D) Bijective

Q18. The set {(a, b)} is called:

(A) Infinite set (B) Set with two elements

(C) Singleton set (D) Empty set

Q19. Imaginary part of 1/(1 + i) is:

(A) 1 (B) 1/2

(C) i/2 (D) -i/2

Q20. √(-5) belongs to the set of:

(A) Rational numbers (B) Real numbers

(C) Complex numbers (D) Integers

Mathematics (Subjective) Group - II

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 1110-XI124

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Let $z \in C$, show that $\bar{\bar{z}} = z$

(ii) Find multiplicative inverse of 1 - 2i

(iii) Simplify: (2, 6) . (3, 7)

(iv) State and prove Golden Rule of fractions.

(v) Find inverse of the given relation $R = \{(x, y) \mid y^2 = 4ax, x \geq 0\}$

(vi) If a, b are elements of a group G, then show that $(ab)^{-1} = b^{-1}a^{-1}$

(vii) If A = [i 0 ; 1 -i], show that $A^4 = I_2$

(viii) If A = [2 3 -2 ; -1 1 5], B = [2 -3 1 ; 5 4 -1] then solve 3X - 2A = B for X.

(ix) Define minor and cofactor of an element of a matrix.

(x) Solve $x^2 + 7x + 12 = 0$ by factorization.

(xi) Discuss the nature of the roots of equation $2x^2 + 5x + 1 = 0$

(xii) Find two consecutive numbers, whose product is 132.

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Define rational fraction.

(ii) Resolve into partial fractions $(7x + 25)/((x + 4)(x + 3))$

(iii) Define arithmetic progression.

(iv) Find three A.Ms between $\sqrt{2}$ and $3\sqrt{2}$

(v) Sum the series: $2 + (1 - i) + (1/i) + \ldots$ to 8 terms.

(vi) Find 9th term of the H.P 1/3, 1/5, 1/7, ...

(vii) Evaluate: 8!/(4!2!)

(viii) Define sample space.

(ix) Find the number of diagonals of a 6-sided figure.

(x) Show that $5^n - 2^n$ is divisible by 3 for n = 4

(xi) Find the middle term of $(1/x - x^2/2)^{12}$

(xii) Expand up to 3 terms (1 - 2x)^(1/3)

Q4. (Marks 18) Attempt any NINE parts:

(i) Define radian.

(ii) Express the angle 75°6′30″ in radian measure.

(iii) Without using the tables, write the values of cos315° and sin540°

(iv) If α, β and γ are the angles of a triangle ABC, then prove that $\tan(\alpha+\beta) + \tan\gamma = 0$

(v) Prove that $\sin(\alpha + \beta)\sin(\alpha - \beta) = \sin^2\alpha - \sin^2\beta$

(vi) Write the domain and range of sinθ

(vii) Find the period of tan(x/3)

(viii) Draw the graph of y = sinx from 0 to π

(ix) At the top of a cliff 80m high, the angle of depression of a boat is 12°. How far is the boat from the cliff?

(x) State any two Law of Cosines.

(xi) Find the area of triangle ABC, in which b = 21.6, c = 30.2 and $\alpha = 52°40'$

(xii) Find the value of $\sin(\cos^{-1}(\sqrt{3}/2))$

(xiii) Define trigonometric equation. Give an example.

> Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5.

Q6.

Q7.

Q8.

Q9.